

Autodesk® Scaleform®

Scaleform

DirectX 9

Scaleform

3D

◦

Ben Mowery
: 3.09
: 2013 10 7

Copyright Notice

Autodesk® Scaleform® 4.3

© 2013 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

Scaleform 4.3

Scaleform Corporation
6305 Ivy Lane, Suite 310
Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

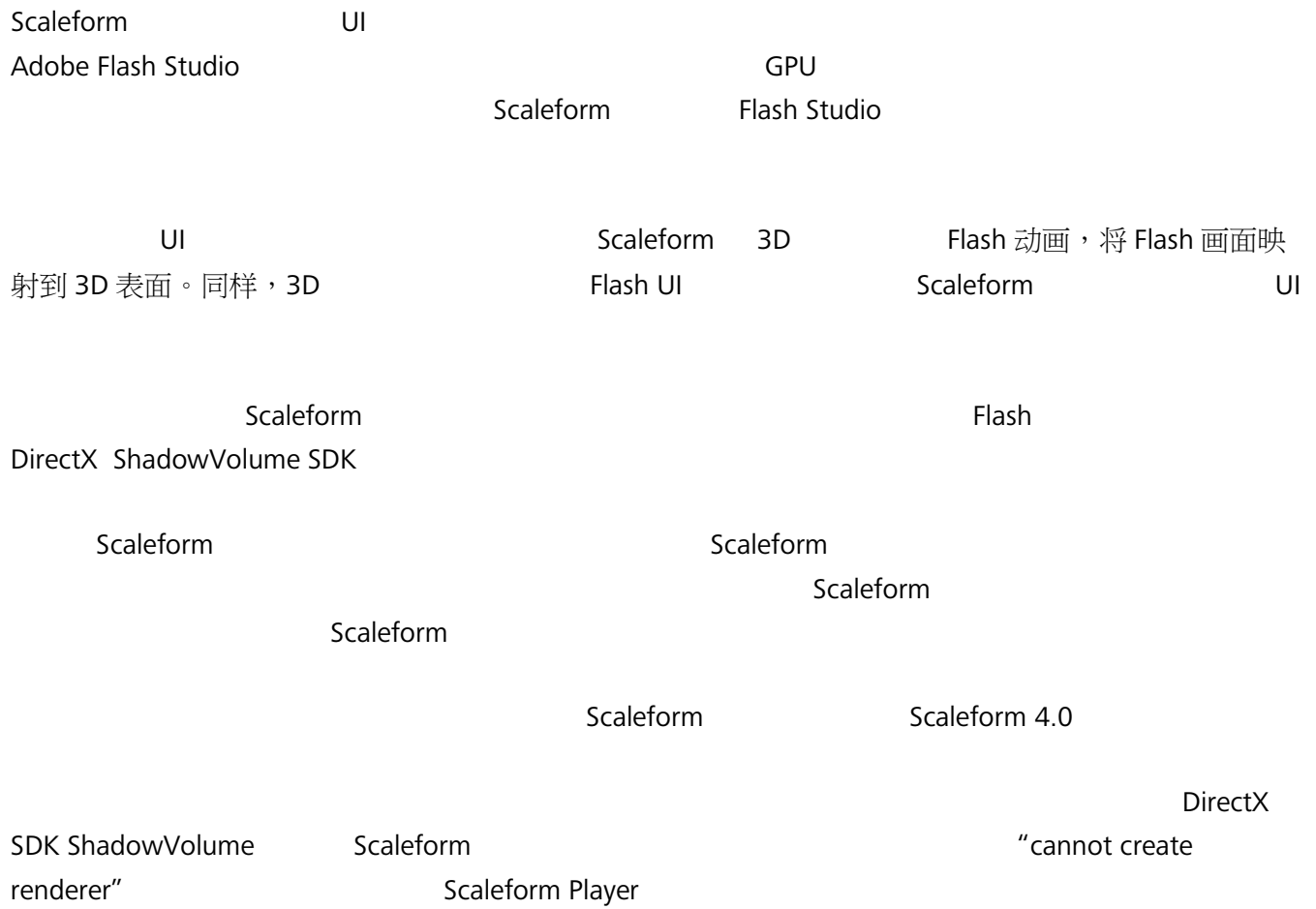
(301) 446-3200

(301) 446-3199

1		1
2		2
3	组建安装和关联编译	3
3.1		3
3.2		4
3.3	Scaleform Player SWF 	4
3.4		6
3.5	Scaleform 	7
3.5.1	AS3_Global.h and Obj\AS3_Obj_Global.xxx 文件	8
4		9
4.1	Flash 	9
4.2		16
4.3		18
4.3.1		18
4.3.2		19
4.3.3	点击测试	21
4.3.4		22
4.3.5		22
5		24
5.1		24
6	C++、Flash	27
6.1	C++.....	27
6.1.1	FSCommand	27
6.1.2	ExternalInterface	29
6.2	C++ 	32
6.2.1		32
6.2.2		34
6.3	多 Flash 文件之间通信.....	35

7	渲染到纹理	38
8	GFxExport	40
9		42

1



2

Scaleform

Scaleform

<https://gameware.autodesk.com/scaleform/developer/>

- Web Scaleform SDK

- PDF

-

Scaleform C++ API

- XML Scaleform XML

- Scale9 Grid Scale9Grid

- IME Scaleform IME

Flash

IME

- Scaleform

3 组建安装和关联编译

3.1

Windows Scaleform
Scaleform installer DirectX Scaleform C:\Program
Files\Scaleform\GFx SDK 4.3

3rdParty

Scaleform libjpeg zlib 。

Projects

Visual Studio 。

App\Samples

Apps\Tutorial

Bin

包含了已编译的演示二进制文档和 Flash 实例：

FxPlayer: Flash HUD 。

IME: IME Flash 。

Samples: Flash FLA

Video Demo Scaleform

Win32: Scaleform Player

AMP: 用于运行 AMP 工具的 Flash 文档 。

GFxExport: GFxExport Flash

7

Include

Scaleform 便利头文 。

Lib

Scaleform 。

Resources

CLIK

Doc

2

PDF

◦

3.2

Scaleform

C:\Program Files

(x86)\Scaleform\GFx SDK 4.3\Projects\Win32\Msvc90\Demos\GFx 4.3 Demos.sln

Demos.sln

Visual Studio

.sln

D3D9_Debug_Static

Scaleform Player

◦

GFx SDK 4.3\Bin\Win32\Msvc90\GFxPlayer\GFxPlayer_D3D9_Debug_Static.exe

Start → All Programs → Scaleform → GFx SDK 4.3 → Demo GFx Player

D3Dx

SWF

Scaleform Flash

◦

3.3 Scaleform Player SWF

Start → All Programs → Scaleform → GFx SDK 4.3 → GFx Players → GFx Player D3D9

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Samples\SWFToTexture

3DWindow.swf

◦



1 : Flash

F1

- | | | | |
|----|--------|-----------|-------|
| 1. | CTRL+F | FPS | ◦ |
| 2. | CTRL+W | Scaleform | Flash |
| | | CTRL+W , | ◦ |
| 3. | | CTRL | ◦ |
| 4. | | CTRL | |
| 5. | CTRL+Z | | |
| 6. | CTRL+U | | |
| 7. | F2 | | |

Flash

800x600 1600x1200

Scaleform

“Scaleform”

Flash

Scaleform

“3D Scaleform Logo”

CTRL+W

CTRL+A

(EdgeAA)

(FSAA)
 AA Scaleform EdgeAA

DP

EdgeAA

Scaleform Player Flash CPU CPU

Scaleform CTRL+Y

60 CPU

release (debug)

Scaleform (debug) Scaleform

3.4

Scaleform 4.3 Scaleform\GFx SDK 4.3\Apps\Tutorial SLN

Scaleform→ GFx SDK 4.3→ Tutorial Windows for Visual Studio 2005 Visual

Studio 2008 “Debug”

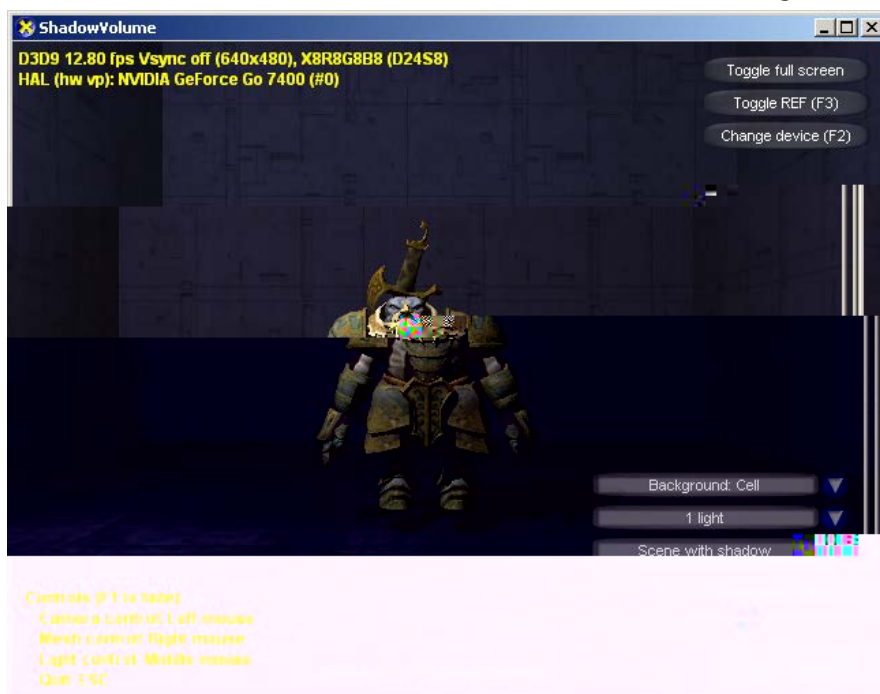


图 2：默认用户界面下的 ShadowVolume 应用画面

3.5 Scaleform

Scaleform	Visual Studio		Tutorial
		\$(GFXSDK)	SDK
	C:\Program Files\Scaleform\GFx SDK 4.3\		
\$(GFXSDK)			"Msvc90 "
Visual Studio 2008	Visual Studio 2010	Msvc90	Msvc10
Scaleform			
\$(GFXSDK)\Src			
\$(GFXSDK)\Include			
	Visual Studio	"Additional Include Directories"	
AS3	AS3		"GFx/AS3/AS3_Global.h"
	AS3		<u>Scaleform LITE</u>

The following library directories should be added to the "Additional Library Directories" field for all build configurations:

```
"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\3rdParty\expat-2.1.0\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\pcre\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\zlib-1.2.7\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\libpng-1.5.13\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\curl-7.29.0\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\jpeg-8d\lib\$(PlatformName)\Msvc90\Debug";
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\$(ConfigurationName)"
```

Msvc90

Visual Studio

Scaleform

```
libgfx.lib
libgfx_as2.lib
libgfx_as3.lib
libgfx_air.lib
libgfxexpat.lib
libjpeg.lib
zlib.lib
libcurl.lib
```

libpng.lib
libgfxrender_d3d9.lib
libgfxsound_fmod.lib

.vcproj

Scaleform include

Tutorial\Section3.5

3.5.1 AS3_Global.h and ObjAS3_Obj_Global.xxx 文件

AS3_Global.h ObjAS3_Obj_global.xxx

AS3_Obj_Global.xxx " "ActionScript 3 swf
" "(script) GlobalObjectCPP C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++
ClassRegistrationTable
AS3_Global.h

ClassRegistrationTable include
ClassRegistrationTable AS3_Global.h

AS3 VM
" "

4

Scaleform 3D Unreal® Engine 3, Gamebryo™, Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™, 和 Trinigy Vision™
Scaleform

Scaleform DirectX DirectX ShadowVolume SDK
DXUT 2D 3D
Scaleform Flash Scaleform DXUT

DXUT
Win32 DirectX Scaleform SDK GFxPlayerTiny.cpp

4.1 Flash

Flash 3D [GFx::Loader](#)
Flash [GFx::MovieDef](#) Flash [GFx::Movie](#)
[Render::Renderer2D](#) [Render::HAL](#) DirectX
Scaleform API
/
ShadowVolume 4.1

步骤 #1: 添加头文件

ShadowVolume.cpp

```
#include "GFx_Kernel.h"  
#include "GFx.h"  
#include "GFx_Renderer_D3D9.h"
```

Scaleform GFxTutorial
Scaleform

Scaleform

```
//          Gfx::Loader
Loader      gfxLoader;

//  SWF/GFx          Gfx::MovieDef
Ptr<MovieDef>  pUIMovieDef;

//          Gfx::Movie
Ptr<Movie>  pUIMovie;

//  Renderer
Ptr<Render::D3D9::HAL>  pRenderHAL;
Ptr<Render::Renderer2D>  pRenderer;
MovieDisplayHandle      hMovieDisplay;
```

#2 Gfx::System

Scaleform

[Gfx::System](#)

Scaleform

WinMain

```
//          Gfx::System
Gfx::System      gfxInit;
```

Gfx::System

Scaleform

Scaleform

WinMain

Gfx::System

Scaleform

Tutorial

Gfx::System

Gfx::System

GfxTutorial

Gfx::System

Gfx::System

Gfx::System::Init() Gfx::System::Destroy()

#3:

Scaleform

WinMain

InitApp()

InitApp()

```
gfx = new GfxTutorial();
assert(gfx != NULL);
```


Gfx::Loader
Gfx::MovieDef

[Gfx::MovieDef](#)

```
// 导入动画
pUIMovieDef = *gfxLoader.CreateMovie(UIMOVIE_FILENAME,
                                       Loader::LoadKeepBindData |
                                       Loader::LoadWaitFrame1, 0);
```

LoadKeepBindData

D3D

LoadWaitFrame1 [CreateMovie](#)
Gfx::ThreadedTaskManager

最后一个参数是可选的，并说明了所用的内存区。有关创建和使用内存区的信息，请参阅 [Memory System Overview](#) 。

#5 :

Gfx::MovieDef

[Gfx::Movie](#)

Gfx::Movie

```
pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);
```

[CreateInstance](#)

false

Flash

最后一个参数是可选

的，并说明了所用的内存区。有关创建和使用内存区的信息，请参阅 [Memory System Overview](#) 。

Advance()

false

CreateInstance

```
// 指向动画第一帧
pUIMovie->Advance(0.0f, 0, true);

// 记录当前时间，测算当前帧到下一帧的时间消耗
MovieLastTime = timeGetTime();
```

Advance

3D

```
pUIMovie->SetBackgroundAlpha(0.0f);
```

Flash

3D

#6 :

Scaleform 在渲染时必须从 Render::D3D9::HAL 获取 DirectX 设备的句柄和参数描述信息。在 D3D 设备创建和 Scaleform 被要求执行渲染前需要调用 Render::D3D9::HAL::InitHAL 函数。

/

D3D

InitHAL

ShadowVolume

OnResetDevice

DXUT

GfxTutorial

OnResetDevice

```
pRenderHAL->InitHAL(
```

```
    Render::D3D9::HALInitParams(pd3dDevice, presentParams,
```

```
    Render::D3D9::HALConfig_NoSceneCalls));
```

InitHAL()函数的调用传递 D3D 设备和参数描述信息到 Scaleform。 HAL_NoSceneCalls

Scaleform

DirectX

BeginScene()

EndScene()

ShadowVolume

OnFrameRender

#7 :

D3D

D3D

ShadowVolume

OnLostDevice

Render::HAL

GfxTutorial

OnLostDevice

D3D

```
pRenderHAL->ShutdownHAL();
```

DXUT

Win32/DirectX

Scaleform SDK

GfxPlayerTiny.cpp

#8 :

Scaleform
WinMain

GfxTutorial
GfxTutorial

```
delete gfx;  
gfx = NULL;
```

DirectX 9

InitHAL() Scaleform
ShutdownHAL()

DirectX 9

InitHAL
InitHAL

D3DPOOL_DEFAULT

D3DPOOL_DEFAULT

ShutdownHAL

D3DPOOL_DEFAULT

DXUT

ShadowVolume ,

DXUT

OnResetDevice

D3DPOOL_DEFAULT

OnLostDevice

GfxTutorial::OnLostDevice

ShutdownHAL

GfxTutorial::OnResetDevice

InitHAL

InitHAL

ShutdownHAL

D3DPOOL_DEFAULT

#9 :

D3D

GfxTutorial::OnResetDevice

// 使用窗口客户端大小尺寸作为视窗大小。

```
RECT windowRect = DXUTGetWindowClientRect();  
DWORD windowWidth = windowRect.right - windowRect.left;  
DWORD windowHeight = windowRect.bottom - windowRect.top;  
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,  
                        windowWidth, windowHeight);
```

[SetViewport](#)

PC

Scaleform

OpenGL

#10 : DirectX

ShadowVolume OnFrameRender() D3D
 BeginScene() EndScene() EndScene()
 GfxTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
    DWORD mtime = timeGetTime();
    float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
    MovieLastTime = mtime;

    pUIMovie->Advance(deltaTime, 0);
    pRenderer->BeginFrame();

    if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
    {
        pRenderer->Display(hMovieDisplay);
    }

    pRenderer->EndFrame();
}
```

Advance() deltaTime
 Gfx::Movie::Advance ,

#11 :

[Render::Renderer2D::Display](#) DirectX D3D
 D3D
 Render::Renderer2D::Display D3D

Scaleform
 DX9

DX9 GfxTutorial::AdvanceAndRender()

```
// 调用Scaleform前保存DirectX状态信息
g_pStateBlock->Capture();

// 描绘画面帧，推进时间计数
gfx->AdvanceAndRender();

// 恢复DirectX状态信息，避免干扰游戏描绘状态
g_pStateBlock->Apply();
```

#12 UI

DXUT

ShadowVolume.cpp

4.1
DXUT

```
// Disable default UI
...
```

DirectX

Flash



图 3：Scaleform Flash 用户界面的 ShadowVolum 应用程序

4.2

GFx::Movie::SetViewport

Flash

Scaleform

Flash

-
-

4:3

Scaleform

800x600

1600x1200

Scaleform

Scaleform

[Gfx::Movie::SetViewScaleMode](#)

GfxTutorial::InitGfx()

Gfx::Movie

```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```

SetViewScaleMode

SM_NoScale	Flash
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

[SetViewAlignment](#)

SM_NoScale

SM_ShowAll

SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode 和 SetViewAlignment

SetViewAlignment

SetViewScaleMode
Scaleform

SM_NoScale

d3d9guide fla

ActionScript

C++

SetViewScaleMode

SetViewAlignment

ActionScript Stage

(State.scaleMode, Stage.align)

4.3

ShadowVolume

Scaleform

Flash

Flash

[GfX::Movie::HandleEvent](#)

GfX::Event

GfX::Movie ◦

4.3.1

ShadowVolume

MsgProc

Win32

GfXTutorial::ProcessEvent

Scaleform

WM_MOUSEMOVE,

WM_LBUTTONDOWN WM_LBUTTONUP :

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);
    if(pUIMovie)
    {
        if(uMsg == WM_MOUSEMOVE)
        {
            MouseEvent mevent(GfX::Event::MouseMove, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if(pMovieButton && uMsg == WM_LBUTTONDOWN)
        {
            ::SetCapture(hWnd);
            MouseEvent mevent(GfX::Event::MouseDown, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if(pMovieButton && uMsg == WM_LBUTTONUP)
        {
            ::ReleaseCapture();
            MouseEvent mevent(GfX::Event::MouseUp, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
    }
}
```

Scaleform

#1:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width,
                    screen_height, 0);
```

(0, 0)
Scaleform

#2

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width / 4,
                    screen_height / 4, 0);
```

HandleEvent

Scaleform

#3 :

```
movie_width = screen_width / 6;
movie_height = screen_height / 6;
pMovie->SetViewport(screen_width, screen_height,
                    screen_width / 2 - movie_width / 2,
                    screen_height / 2 - movie_height / 2,
                    movie_width, movie_height);
```

(0,0)

(screen_width /

2 - movie_width / 2, screen_height / 2 - movie_height / 2)

Flash

[Gfx::Movie::SetViewAlignment](#)

[Gfx::Movie::SetViewport](#)

SetViewAlignment

[SetViewScaleMode](#)

Scaleform

4.3.2

[Gfx::Movie::HandleEvent](#)

[Gfx::KeyEvent](#)

[Gfx::CharEvent:](#)

```
KeyEvent(EventType eventType = None,
          Key::Code code = Key::None,
          UByte asciiCode = 0,
          UInt32 wcharCode = 0,
          UInt8 keyboardIndex = 0)
```

```
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```


Gfx::KeyEvent Gfx::CharEvent ASCII Windows
 Gfx::CharEvent WM_CHAR Gfx::KeyEvents WM_SYSKEYDOWN,
 WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP

- 'c' SHIFT : Gfx::KeyEvent WM_KEYDOWN
 - 'c'
 -
 - SHIFT
- Gfx::CharEvent WM_CHAR “ ” ASCII 'C'
 Gfx ◦
- 'c' Gfx::KeyEvent WM_KEYUP ◦
 Gfx::CharEvent ◦
- F5 Gfx::KeyEvent
 F5 ASCII Gfx::CharEvent

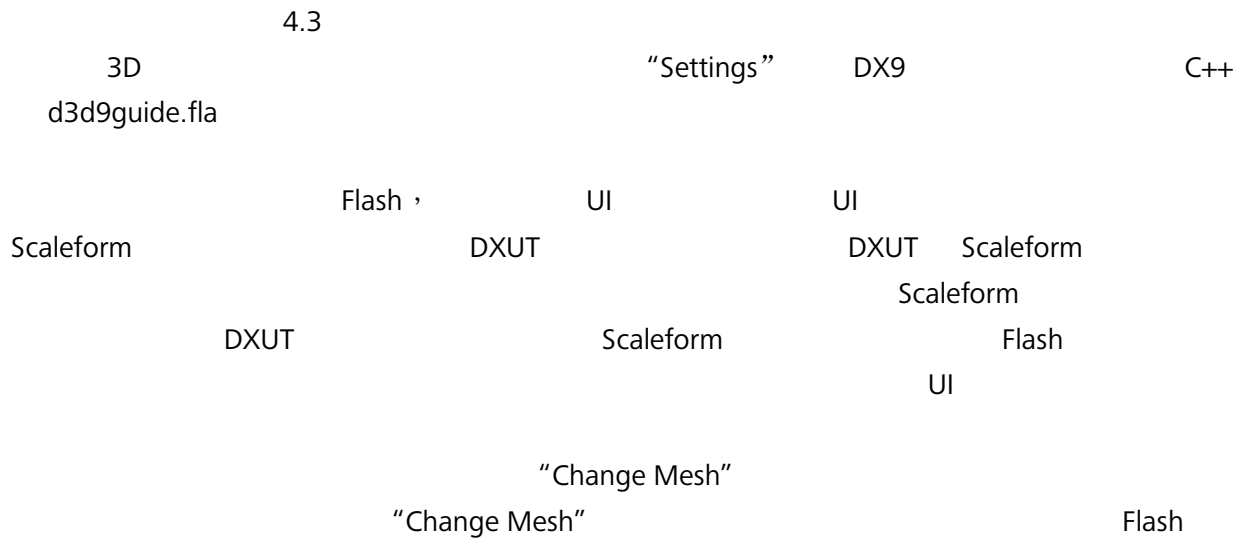
 Gfx::KeyEvent Gfx_Event.h
 Flash GfxPlayerTiny.cpp Scaleform Player
 Windows Flash ProcessKeyEvent
 3D

```

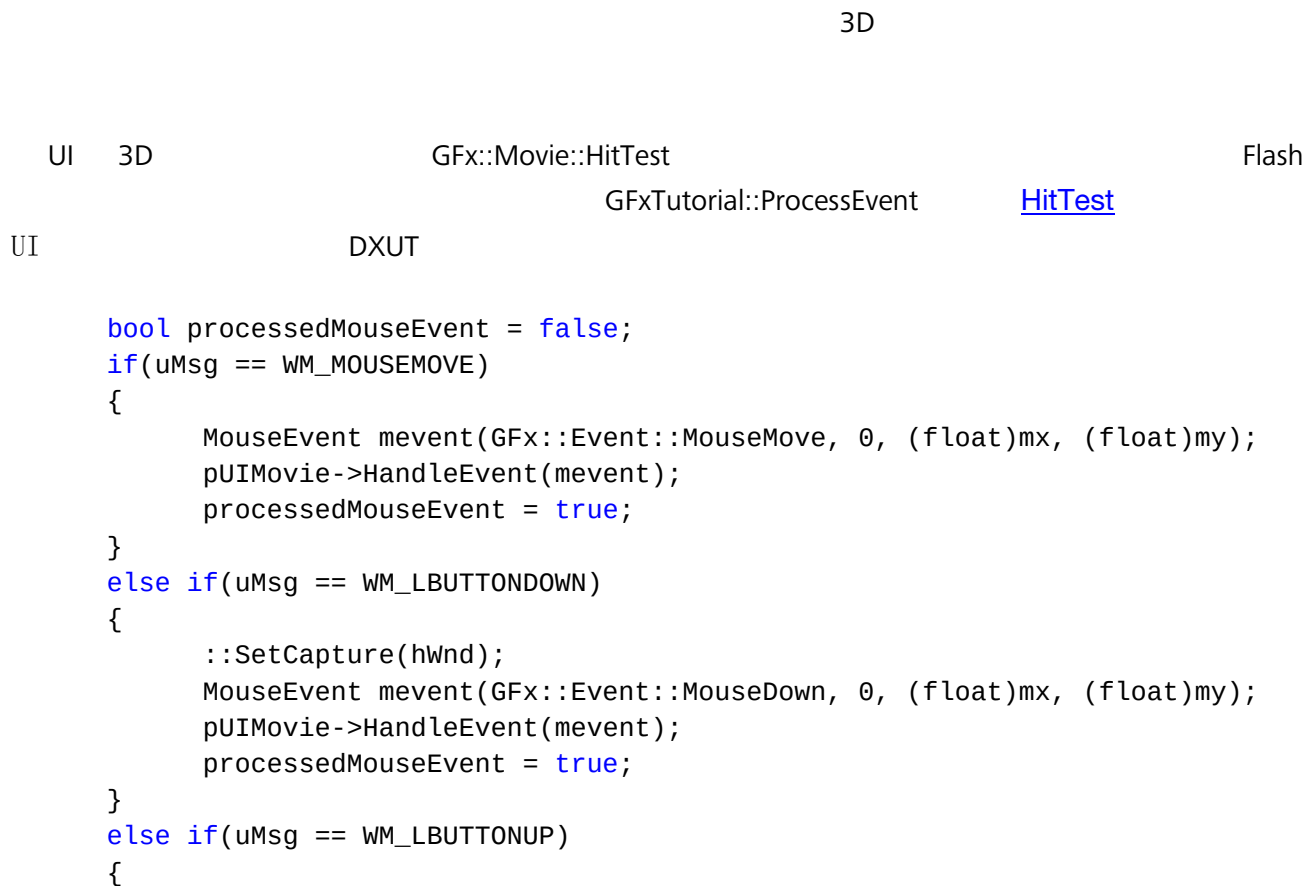
void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM lParam)
  
```

Windows WndProc WM_CHAR, WM_SYSKEYDOWN, WM_SYSKEYUP,
 WM_KEYDOWN WM_KEYUP Scaleform pMovie

 Gfx::KeyEvent Gfx::CharEvent Gfx::CharEvents
 ASCII Unicode
 IME . IME IME
 Gfx::KeyEvent Page Up Page Down



4.3.3 点击测试



```

        ::ReleaseCapture();
        MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
        pUIMovie->HandleEvent(mevent);
        processedMouseEvent = true;
    }

    if(processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
        Movie::HitTest_Shapes))
        *pbNoFurtherProcessing = true;

```

4.3.4

“Change Mesh”

4.3.2 W,S,A,D 和 Q 3D
Scaleform 3D

6.1.2

C++

4.3.5

触摸事件就像鼠标事件一样，但只在用户正触摸屏幕时才被发送。要利用触摸事件，我们可以将原始触摸数据从 Windows API 直接路由到 Scaleform 之中。要求的最低平台是在具有触摸功能的系统上运行的 Windows 7，如果不能满足此要求，就无法继续学习本教程的这一节内容，而是应该跳到下一节。

编译之前，将 **WINVER=0x601** 添加到项目的“预处理器定义”。这就指示 Windows API 编译针对 Windows 7 的功能（即多点触控）。在 ShadowVolume.cpp 中，取消宏 **#define ENABLE_MULTITOUCH** 的注释，此宏可防止编译不受支持的代码。

为了使用特殊的多点触控库，添加了 **#include <windows.h>**。在窗口初始化期间，我们通过 RegisterTouchWindow 注册我们的侦听触摸事件的意图，而 RegisterTouchWindow 允许接收 MsgProc 回调中的 WM_TOUCH 消息。以便进行处理。

```

    if(uMsg == WM_TOUCH)
    {
        ProcessTouchEvent(pUIMovie, uMsg, wParam, lParam);
    }

```

ProcessTouchEvent 是负责解析和传递 WM_TOUCH 消息的函数。Windows 函数用来将触摸坐标转换为像素指标。然后，创建一个 GFx::TouchEvent，并通过 HandleEvent 将其发送到电影。GFx::TouchEvent 类包含一个唯一 ID（由 OS 管理）、x/y 坐标、接触面积、压力（从 0..1）以及它是不是一个主触点。

```

TouchEvent( EventType evtType,
             unsigned id,
             float _x,
             float _y,
             float wcontact = 0,
             float hcontact = 0,
             bool primary = true,
             float pressure = 1.0f)

```

SWF 电影现在正在接收触摸事件，但默认情况下，GfX 最多支持 0 个触点。最后一步是定义一个继承来的 MultitouchInterface，用来描述我们的多点触控环境，具体说来就是支持的最大触点数，以及我们是否也将传递 GfX::GestureEvent（本教程不包含本机手势事件，但您可以使用本节概述的原理，以将它们公开到 Scaleform）。

```

class FxPlayerMultitouchInterface : public MultitouchInterface
{
public:
    // 返回硬件支持的最大触点数
    virtual unsigned GetMaxTouchPoints() const { return 2; }

    // 返回支持的手势的位掩码（此时什么也没有）
    virtual UInt32    GetSupportedGesturesMask() const { return 0; }

    // 支持多点触控？
    virtual bool      SetMultitouchInputMode(MultitouchInputMode) { return
true; }
};

```

一旦我们通过 SetMultitouchInterface 将此接口类的一个引用发送到我们的电影，就成功地将触摸事件传播到了该电影。

5

5.1

Scaleform

TTF

Flash

Windows Linux

Scaleform

<https://gameware.autodesk.com/scaleform/developer/?action=doc>。

5000

Scaleform

Scaleform
Scaleform

“ ” “ ”

TTF

Scaleform

Bin\Data\AS2\Samples\FontConfig
Player D3D9

sample.swf

Scaleform

早上好

你好!早上好!

多想你呀!

Chinese (繁體中文版)

///

///|///|

///|

图 3a：中文小字体

///

图 4b：线状描绘

CTRL+W



图 4c：大号中文字

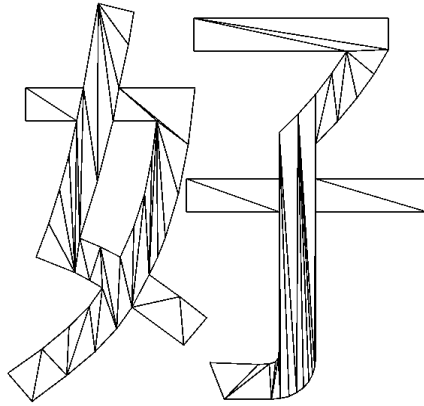


图 4d：线状描绘

4c

4d

Scaleform

Flash

gfx_2.1_texteffects_sample.zip



图 5：文本效果

6 C++ 、Flash

Flash

Flash

Scaleform

C++

Scaleform

C++

6.1

C++

Scaleform

C++

: FSCCommand

ExternalInterface

FSCCommand 和 ExternalInterface

GfX::Loader

C++

FSCCommand

fscommand

ExternalInterface

flash.external.ExternalInterface.call

[GfX::Value](#)

6.1.2

FSCCommands

ExternalInterface

fscommands

gfxexport

SWF

fscommands

-fstree

-fslist

-fsparams

ExternalInterface

fscommands

6.1.1 FSCCommand

fscommand

fscommand("setMode", "2");

fscommand

ExternalInterface

GfX FSCCommand

[GfX::FSCCommandHandler](#)

FSCCommand

GfX::Loader

GfX::Movie

GfX::Movie ,

fscommand

Scaleform PlayerTiny

("FxPlayerFSCCommandHandler") ,

ShadowVolume 。

6.1

Gfx::FSCommandHandler

```
class OurFSCommandHandler : public FSCommandHandler
{
public:
    virtual void Callback(Movie* pmovie,
                          const char* pcommand, const char* parg)
    {
        GfxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

Callback

fscommand

fscommand

GfxTutorial::InitGfx()

Gfx::Loader

// 注册 FSCommand 句柄

```
Ptr<FSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);
```

Gfx::Loader

Gfx::Movie

◦ SetFSCommandHandler

fscommand

ShadowVolume

"Toggle Fullscreen"

FSCommand: ToggleFullscreen, Args:

Open Flash/HUDMgr.as in Flash Studio. This class is referenced from d3d9guideAS3.fla. The association between this external class and the Flash content is made by setting the ActionScript class for the "hudMgr" Symbol in d3d9guideAS3's Symbol Library.

Compare the events printed to the screen with the fscommand() calls made from the HUDMgr class in the following function:

```
function toggleFullscreen(ev:MouseEvent) {
    fscommand("ToggleFullscreen");
}
```

As an exercise, change fscommand("ToggleFullscreen") to fscommand("ToggleFullscreen", String(hud.visible)) to print the value of the hud.visible value to C++. Export the flash movie (CTRL+ALT+Shift+S) and replace d3d9guideAS3.swf.

动态脚本 fscommand()调用

FSCommand

ShadowVolume

fscommand

```
if(strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

DXUT

OnFrameMove

OnFrameMove

```
if(doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullScreen();
}
```

fscommand

doToggleFullscreen

DXUT

OnFrameMove

6.1.2 ExternalInterface

Flash ExternalInterface

fscommand

ExternalInterface

fscommand

```
class OurExternalInterfaceHandler : public ExternalInterface
{
public:
    virtual void Callback(Movie* pmovieView,
                          const char* methodName,
                          const Value* args,
                          unsigned argCount)
    {
        GfxPrintf("ExternalInterface: %s, %d args: ",
                  methodName, argCount);
        for(unsigned i = 0; i < argCount; i++)
        {
            switch(args[i].GetType())
```

```

        {
            case Value::VT_Null:
                GfxPrintf("NULL");
                break;
            case Value::VT_Boolean:
                GfxPrintf("%s", args[i].GetBool()? "true" : "false");
                break;
            case Value::VT_Int:
                GfxPrintf("%s", args[i].GetInt());
                break;
            case Value::VT_Number:
                GfxPrintf("%3.3f", args[i].GetNumber());
                break;
            case Value::VT_String:
                GfxPrintf("%s", args[i].GetString());
                break;
            default:;
                GfxPrintf("unknown");
                break;
        }
        GfxPrintf("%s", (i == argCount - 1) ? "" : ", ");
    }
    GfxPrintf("\n");
}
};

```

Gfx::Loader

```

Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
gfxLoader.SetExternalInterface(pEIHandler);

```

Open d3d9HUD.as used by

d3d9guideAS3.fla and look at the ActionScript for the following functions:

```

function onSubmit(path:String) {
    ExternalInterface.call("MeshPath", path);
    CloseMeshPath();
}

function onFocusIn(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", true);
}

function onFocusOut(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", false);
}

```

MeshPathFocus 当文本输入框获得或者失去焦点时被调用。

(true false)

“MeshPathFocus”

```
Callback! MeshPathFocus, nargs = 1
    arg(0) = true
```

```
Callback! MeshPathFocus, nargs = 1
    arg(0) = false
```

GfXTutorial

```
if(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1 &&
args[0].GetType() == Value::VT_Boolean) {
    if (args[0].GetType() == Value::VT_Boolean)
        gfx->SetTextboxFocus(args[0].GetBool());
}
```

GfXTutorial::ProcessEvent

3D

```
if(uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
    uMsg == WM_KEYDOWN || uMsg == WM_KEYUP ||
    uMsg == WM_CHAR)
{
    if(textboxHasFocus || wParam == 32 || wParam == 9)
    {
        ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
        *pbNoFurtherProcessing = true;
    }
}
```

(ASCII 32) tab(ASCII 9) “Toggle UI” “Settings”

“Change Mesh”

OurExternalInterfaceHandler::Callback

```
static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";
```

```

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
    doChangeMesh = true;
    const char *filename = args[0].GetString();
    MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
        changeMeshFilename, _countof(changeMeshFilename));
}

DXUT OnFrameMove DXUT

```

6.2 C++

6.1 C++ Scaleform
C++ Scaleform C++

6.2.1

Scaleform [GetVariable](#) [SetVariable](#) 6.2
HUD (fxplayer.swf) Scaleform Player F5

```

void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
    LPARAM lParam, bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);

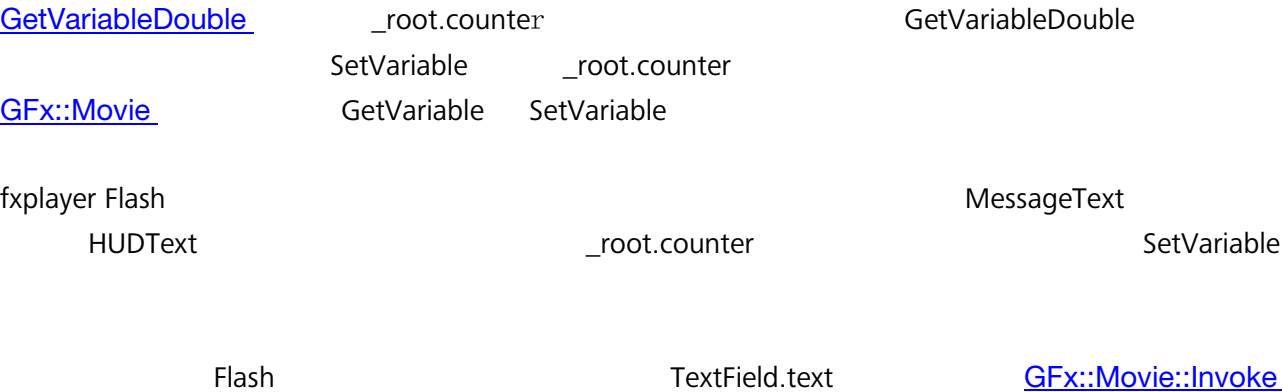
    if(pHUDMovie && uMsg == WM_KEYDOWN)
    {
        if(wParam == VK_F5)
        {
            int counter = (int)pHUDMovie->GetVariableDouble("_root.counter");
            counter++;
            pHUDMovie->SetVariable("_root.counter",
                Value((double)counter));
            char str[256];
            sprintf_s(str, "testing! counter = %d", counter);
            pHUDMovie->SetVariable("_root.MessageText.text", str);
        }
    }
}

```

```

    }
}
...

```



6.2.2



图 8：SetVariable 改变 HUD 文本值。



SV_Normal)	1	SV_Sticky
	3 _root.mytextfield.text	C++

SetVariableArray	Flash
GfxTutorial::InitGfx()	_root.SceneData

```
// 初始化场景
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
    "_root.SceneData", 0, sceneData, 3);
```

6.1
ShadowVolume

C++
C++

6.2.2

[Gfx::Movie::Invoke](#)

UI

Gfx::Movie::Invoke	"Change Mesh"	F6 F7
		SetVariable
SetVariable	Invoke ActionScript	
Gfx::Movie::Invoke	WM_CHAR	openMeshPath
...		
<pre>else if (wParam == VK_F6) { bool retval = pHUDMovie->Invoke("root.ui.hud.OpenMeshPath", ""); GfxPrintf("root.ui.hud.OpenMeshPath returns '%d'\n", (int) retval); }</pre>		

```

else if (wParam == VK_F7)
{
    const char *retval = pHUDMovie->Invoke(
        "root.ui.hud.CloseMeshPath", "");
    GfxPrintf("root.ui.hud.CloseMeshPath returns '%d'\n", (int) retval);
}

```

Scaleform

[Gfx::Movie::Advance](#) [Gfx::MovieDef::CreateInstance](#)
 initFirstFrame true 1
 printf Gfx::Value
[InvokeArgs](#) va_list Invoke
 InvokeArgs printf vprintf

6.3 多Flash 文件之间通信

C++ C++ SWF
 C++
 MMOG HUD HUD
 HUD HUD
 SWF C++
 Gfx::Movie
 SWF Gfx::Movie loadMovie unloadMovie
 C++ SWF Gfx::Movie
 container.swf ActionScript container.swf
 MovieClipLoader ActionScripts

```

var mclLoader:MovieClipLoader = new MovieClipLoader();

```



```

    ActionScript
function loadMovie(url:String):void {
    var loader = new Loader();
    loader.load( new URLRequest( url ) );
    addChild( loader )
}

```

```

loadMovie
root.inventoryWindow.widget1.counter

```

```

Flash
root
level 6 root.counter level6.counter

```

```

loadMovie
root.myMovie
root.counter
lockroot = true Lockroot ActionScript
root root root ActionScript
Adobe Flash
Gfx::Movie

```

Container fla

```

loadMovie
loadClip
Loader
Container fla
ActionScript ExternalInterface C++

```

```

function loadMovie(url:String):void {
    var loader = new Loader();
    loader.load( new URLRequest( url ) );
    loader.contentLoaderInfo.addEventListener( Event.OPEN, handleLoadOpen,
                                                false, 0, true );
    loader.contentLoaderInfo.addEventListener( Event.INIT, handleLoadInit,
                                                false, 0, true );
    loader.contentLoaderInfo.addEventListener( Event.PROGRESS,
                                                handleLoadProgress, false, 0, true );
    loader.contentLoaderInfo.addEventListener( Event.COMPLETE,

```

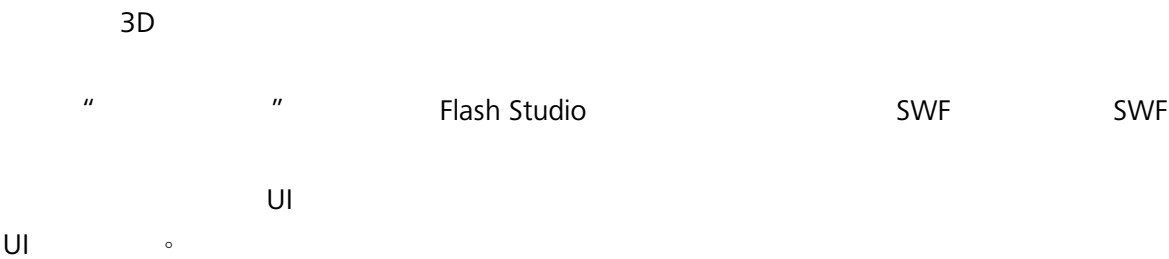
```

                                handleLoadComplete, false, 0, true );
    addChild( loader )
}

function handleLoadOpen( e:Event ):void {
    trace("Event.OPEN - Load Started!");
}
function handleLoadInit( e:Event ):void {
    trace("Event.INIT - Loaded Content Ready!");
}
function handleLoadProgress( e:Event ):void {
    trace("Event.PROGRESS - Load Progress!");
}
function handleLoadComplete( e:Event ):void {
    trace("Event.COMPLETE - Load Complete!");
}

```

7 渲染到纹理



1. cell.x (cellwall.jpg)

```
cellwallIRT.jpg MeshMaterialList
Material {
1.000000;1.000000;1.000000;1.000000;;
40.000000;
1.000000;1.000000;1.000000;;
0.000000;0.000000;0.000000;;
```

```
TextureFilename {
"cellwallRT.jpg";
}
```

2. cell.x DXUT

DXUT

CreateTexture

(Render

Texture)

DXSDK

DXUT

```
pd3dDevice->CreateTexture(rtw,rth,0,
D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
D3DP00L_DEFAULT, &g_pRenderTex, 0);
```

```
g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();
```

3. AdvanceAndRender GFx

a.

```
pd3dDevice->GetRenderTarget(0, &poldSurface);
pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
```

b.

```
ptex->GetSurfaceLevel(0, &psurface);
HRESULThr = pd3dDevice->SetRenderTarget(0, psurface );
```

c.

Display

a

8 OpenGL

我们还包含了一个小的 OpenGL/GLUT 集成样例 (Tutorial\OpenGL)，它基于 Nvidia 曲面细分 (Tessellation) 示例 (<http://developer.download.nvidia.com/SDK/10/opengl/samples.html#tessellation>)。这是一个非常小的样例，它使用一种与 D3D9 教程略有不同的方法。通过比较 tessellation.cpp 与 tessellation_original.cpp 您就会发现，只是在原来的 NVIDIA 样例中添加了几行代码。主程序通过高级调用（如 Init()、ShutDown、AdvanceAndDisplay() 等）与 GfX 通信，而且所有 GfX 专用实现代码都封装到了 GfXPlayerImpl 类 (GfXPlayerGL.cpp) 之中。

9 GfXExport

SWF SWF SWF SWF

SWF

GfXExport SWF

DDS DXT

GfXExport

Scaleform GfXExport

d3d9guide.swf fxplayer.swf Scaleform

```
gfxexport -i DDS -c d3d9guideAS3.swf
gfxexport -i DDS -c fxplayer.swf
```

gfxexport.exe C:\Program Files\Scaleform\\$(GFXSDK)

convert.bat 7

-i DDS DDS DirectX DXT

DXT

-c Scaleform

DXT

-share_images

ShadowVolume

8

Scaleform

Gfx::Loader::CreateMovie

10

Scaleform

- Flash 3 Scaleform Player SWF SDK Gamebryo

- [FAQs](#)

- Scale9 [Scale9Grid Overview](#)

- Scaleform SWF Flash
[Gfx::FileOpener](#)

- [Gfx::ImageCreator](#)

- [Gfx::Translator](#)